

Comparative Insights: M20 Concrete With vs Without Graphene

This document presents a comparative technical analysis of two NABL-accredited test reports for M20 grade concrete, one with Graphene-based admixture and one without Graphene, focusing on mix design parameters and compressive strength performance.

Report Overview

Report A: M20 Concrete WITH Graphene Admixture

Test Report No.: BNR-1127-TR-790403

Testing Period: July–August 2025

Report B: M20 Concrete WITHOUT Graphene Admixture

Test Report No.: BNR-1140-TR-793018

Testing Period: July 2025

Key Mix Design Comparison

Cement Content:

- With Graphene: 331 kg/m³
- Without Graphene: 371 kg/m³

Water–Cement Ratio:

- With Graphene: 0.48
- Without Graphene: 0.55

Workability (Slump):

- With Graphene: 160 mm
- Without Graphene: 130 mm

Compressive Strength Performance

Accelerated Test (Equivalent 28-Day Strength):

- With Graphene: ~25.5 MPa
- Without Graphene: ~25.0 MPa

Actual 28-Day Compressive Strength:

- With Graphene: Average ~29.0 MPa
- Without Graphene: Data limited to accelerated results only

Technical Insights

- Graphene-modified concrete achieved higher compressive strength with ~40 kg/m³ lower cement content.
- Improved workability was observed despite reduced cement dosage.
- Lower water–cement ratio contributed to denser microstructure and strength gain.

- Results indicate improved cement efficiency and performance enhancement due to Graphene addition.

Commercial & Practical Implications

- Potential reduction in cement consumption without compromising strength.
- Improved pumpability and workability benefits site execution.
- Lower cement usage translates to cost and CO₂ footprint reduction.
- Suitable for sustainable and high-performance concrete applications.

Conclusion

The comparative analysis demonstrates that Graphene-enhanced M20 concrete offers measurable performance advantages over conventional M20 concrete. These findings support the technical and commercial viability of Graphene-based admixtures in structural concrete applications.